

Deepfake & AI-Generated Image Detection:

A Comparative Study of Convolutional Neural Networks and Vision Transformer Approaches

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Abstract

The sudden increase of AI-generated images poses significant challenges to information integrity, digital trust, and media authenticity. This paper will go over the details of the development and comparative evaluation of two deep learning approaches for binary classification of real versus AI-generated images. Our primary model, EfficientNetB4 (V3), was trained using a two-phase transfer learning strategy on approximately ~257,283 images from five publicly available datasets, achieving a Test AUC-ROC of 0.9751 and Test Accuracy of 92.20%. A specialized Vision Transformer (ViT-XAI) model, trained sequentially on CIFAKE and 5,000 StyleGAN facial images, achieved an AUC-ROC of 0.91. Both models were deployed as live HuggingFace Spaces applications with visual explainability. Grad-CAM is used for EfficientNetB4 and multi-layer attention maps for ViT. Real-world testing revealed important limitations in both models. V3 fails on images generated by Midjourney because of their absence from its training data, while ViT produces false positives on authentic photographs. Additionally, we developed an ensemble combining both architectures via a logistic regression meta-learner trained on ViT-V1, ViT-V2 (a continual-learning variant), and EfficientNetB4 outputs, achieving 92.25% accuracy on a 2,000-image blind validation set with reduced false-positive rates. This work documents model failures honestly, compares all architectures across multiple dimensions, and proposes concrete next steps for improvement.

1 Introduction

With the rise of generative AI tools, such as GANs, diffusion models, and large-scale synthesis systems (like Midjourney and Stable Diffusion), it has become easily accessible for people to generate photorealistic images that are almost identical to real images. These deepfakes and AI-generated images pose growing threats such as the spread of visual misinformation, the creation of fake digital evidence, identity fraud, and the loss of public trust in digital content.

Automated detection systems are an important line of defense, yet no model has achieved strong generalization across all types of AI images. This project looks into multiple complementary approaches: a CNN based on EfficientNetB4 fine-tuned across a large multi-domain dataset (V3), a Vision Transformer (ViT) model specialized for forensic facial analysis, and an ensemble meta-learner that combines both architectures. Notably, no single model is perfect. V3 fails on Midjourney images not in its training data, and the ViT model yields false positives on real photographs. These restrictions are documented and evaluated throughout.

Our key contributions are: (1) a progressively developed EfficientNetB4 detector across several versions (V1-V4), showing the impacts of dataset diversity and domain-specific fine-tuning, (2) a ViT forensic detector with multi-layer attention visualization, (3) an ensemble meta-learner combining ViT and CNN architectures that achieves 92.25% accuracy with improved false-positive control, (4) deployment of models as live public use, and (5) an honest comparative analysis including real-world failures and lessons learned.

2 Related Work

Initial deepfake detection depended on features made by humans, such as blinking patterns, facial landmarks, and compression artifacts [1]. Deep learning changed the approach by using fully trainable detectors from start to finish. EfficientNetB4, which scales network depth, width, and resolution at the same time, achieves greater accuracy with fewer parameters, making it ideal for transfer learning [2, 3]. Vision Transformers [4], processing images as sequences of 16×16 patches, capture long-range dependencies that CNN architectures often miss — a property valuable for detecting globally distributed generation artifacts. Grad-CAM [5] provides spatial saliency maps for CNN predictions; multi-head attention visualization serves an analogous role for ViT. The CIFAKE dataset [6], combining real CIFAR-10 images with Stable Diffusion fakes, serves as the primary benchmark for this work. Ensemble methods and meta-learning approaches have shown promise in combining complementary detectors: stacking classifiers trained on different feature representations can improve robustness when individual models exhibit non-overlapping failure modes [7].

3 Data

3.1 EfficientNetB4 (V3) Dataset

V3 was trained on around ~257,283 images from five sources, maintaining an almost perfect class balance. One critical gap in the data is that a planned Midjourney dataset was deleted from HuggingFace before training started. This directly caused the failure documented in Section 5.3.

Dataset	Type	~Images	Source
CIFAKE	Real + Stable Diffusion fakes	100,000	Kaggle
140k Real & Fake Faces	Real + GAN-generated faces	140,000	Kaggle
OpenForensics	Face manipulation dataset	10,000	Kaggle
Natural Images HD	Real photographs	5,000	Kaggle
HEMG / FakeShield	Mixed real/AI images	25,000	HuggingFace
Total	Balanced real/fake	~257,283	Combined

All images were resized to 224x224 pixels, normalized to float32, and split 90/10 into training (231,554) and validation (25,728) sets. The CIFAKE test set (20,000 images) was reserved for final evaluation. Individual counts reflect raw dataset sizes before merging. Filename overlaps between datasets and removal of empty files during preprocessing reduced the final count to 257,283 images (128,582 real / 128,701 fake).

3.2 Vision Transformer (ViT) Dataset

To use the ViT data, there had to be two stages. Stage 1 pre-trained on the full CIFAKE dataset for general AI artifact detection. Stage 2 fine-tuned 5,000 high-resolution facial images (2,500 real human faces and 2,500 StyleGAN-generated faces) for forensic face manipulation specialization. All images were standardized to 224x224x3 pixels.

3.3 Ensemble Datasets

Dataset	Purpose	Images	Source	Used By
CIFAKE + Manjil Karki Deepfake and Real Images	ViT-V2 continual learning training	50,000	Kaggle	ViT-V2
Kshitiz Bhargava Deepfake Face Images	Simple ensemble evaluation	600 (sampled)	Kaggle	Simple Ensemble
Manual Curated Set	Meta-learner calibration	67	Hand-labeled	Meta-Learner
Manjil Karki Deepfake and Real Images (validation split)	Ensemble blind validation	2,000	Kaggle	ViT-V2

ViT-V2 Continual Learning Training Set: 50,000 images (25,000 real, 25,000 fake) drawn from CIFAKE, the Manjil Karki Deepfake, and Real Images dataset (Kaggle). The Manjil Karki set contains GAN-generated face swaps with real CelebA-sourced faces. We used an 80/20 train/validation split (40,000 training, 10,000 validation) layered by class. Data augmentation included random horizontal flips, 10° rotation, and color jitter (brightness/contrast ± 0.2), with ImageNet normalization applied to all inputs. A later fine-tuning attempt using CIPLAB Real and Fake Face Detection and DALL-E Recognition Dataset images to improve ViT-V2's Midjourney detection resulted in catastrophic forgetting and was abandoned. The pre-fine-tuning checkpoint was used in the final ensemble.

Simple Ensemble Evaluation Set: 600 balanced images (300 real, 300 fake) sampled from the Kshitiz Bhargava Deepfake Face Images dataset, which contains GAN-generated face images. This set of images was used to evaluate the simple ViT + CNN ensemble and was not used for training any model.

Meta-Learner Calibration Set: 67 manually curated and labeled images (33 fake, 34 real) were used to train the logistic regression meta-learner. Each image was run through all three component models (ViT-V1, ViT-V2, CNN) to produce the P(fake) feature vectors. This set was split 80/20 for meta-learner training and testing.

Blind Validation Set: 2,000 images (1,000 real, 1,000 fake) were randomly sampled from the Manjil Karki dataset's validation section. This set was held out from all training and calibration to get an unbiased estimate of the performance.

4 Methods

4.1 V1 — EfficientNetB4 Baseline (Google Colab)

V1 used EfficientNetB4 pre-trained on ImageNet and fine-tuned on CIFAKE alone (~100,000 images) on Google Colab. Training went through two phases. Phase 1 (10 epochs, frozen backbone, Adam lr=1e-3) and phase 2 (25 epochs, top 30 layers unfrozen, lower lr). Custom head: GlobalAveragePooling, BatchNormalization, two Dropout(0.3) layers, Dense(256, ReLU), sigmoid output. Binary Cross-Entropy with label smoothing (0.1). V1 resulted in a Test AUC-ROC of 0.9924 and 96.00% accuracy on CIFAKE. It's strong within this narrow domain, but has poor generalization to other image types.

4.2 V2 — Multi-Dataset Attempt (Google Colab, Crashed — No Notebook)

V2 was made as an attempt to extend V1's training to a five-dataset regime on Google Colab. Colab's session time limits, memory constraints, and limited storage caused the training run to crash repeatedly before convergence on the 257K-image dataset. We switched to Lightning AI (Tesla T4 GPU), which provided stable sessions and sufficient memory. V2 produced no stable checkpoint and has no associated notebook.

4.3 V3 — Multi-Dataset EfficientNetB4 (Lightning AI, Primary Model)

V3 used the same architecture as V1; the only difference is that it was trained on all five datasets on Lightning AI (Tesla T4). This took about 20 hours to run. Same two-phase strategy: Phase 1 (10 epochs, Adam lr=1e-3, EarlyStopping patience=4, ReduceLROnPlateau factor=0.5) and Phase 2 (25 epochs, Cosine Decay lr schedule from 1e-5). V3 was trained in a Lightning AI terminal rather than an interactive notebook. The supplementary notebook was reconstructed from terminal logs. All metrics shown are from the actual live training run.

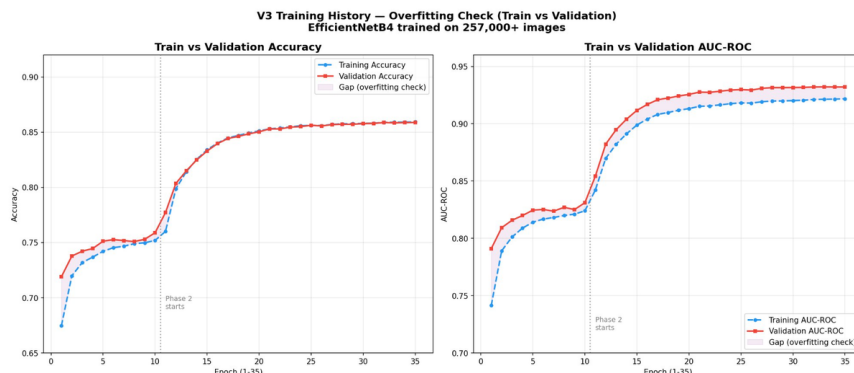


Figure 1: V3 Training History (35 epochs). The dotted line marks Phase 1 to Phase 2 transition. Close train/validation tracking confirms no significant overfitting.

4.4 V4 — Midjourney Fine-Tuning (Attempted, Reverted to V3)

After V3 deployment, real-world testing kept misclassifying Midjourney V5/V6 images as real. A V4 fine-tuning experiment on Lightning AI used 3,000 Midjourney fakes and 8,000 real images for 9 epochs. V4 achieved 99.33% validation accuracy on the Midjourney-specific set, but immediately began classifying real photographs as AI-generated (catastrophic forgetting). Raising the threshold from 0.50 to 0.65 was insufficient. V3 was reinstated as the production model. V4 is documented in the V3 notebook but has no separate notebook, since it was run entirely in a Lightning AI terminal.

4.5 Vision Transformer (ViT-XAI)

Architecture: google/vit-base-patch16-224-in21k — 12 Transformer encoder layers, 768 hidden dimensions, 12 attention heads per layer, around 86 million parameters. Images are divided into 16x16 pixel patches, linearly projected, with an added [CLS] token for classification. Training: AdamW optimizer, Binary Cross-Entropy loss, sequential transfer learning (Stage 1: CIFAKE generalist, Stage 2: StyleGAN face specialist). Training loss converged from 0.807 (step 50) to 0.436 (step 400).

Step	Training Loss
50	0.807476
100	0.722876
150	0.688455
200	0.640010
250	0.593748
300	0.455064
350	0.442232
400	0.435844

Figure 2: ViT training loss log — steady convergence across 400 steps.

Explainability uses a 4-layer forensic attention visualization: Layer 1 (initial texture), Layer 4 (facial geometry), Layer 8 (artifact search), Layer 12 (final decision).

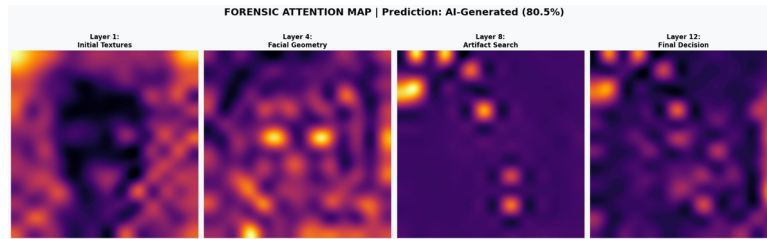


Figure 3: ViT multi-layer attention map for a Midjourney-generated image (80.5% AI confidence). Attention progressively focuses from global textures to specific forensic artifact hotspots.

4.6 Ensemble Model (ViT + EfficientNetB4)

To address the failures of V3 (Midjourney blind spot) and the ViT (false positives on real photos), we developed two ensemble approaches combining both architectures.

Simple Ensemble (ViT + CNN): This ensemble combined the ViT-XAI model and V3 model via equal-weight probability averaging. Each model’s output was converted to $P(\text{fake})$, and the ensemble prediction was computed as: $P_{\text{ensemble}}(\text{fake}) = 0.5 \times P_{\text{ViT}}(\text{fake}) + 0.5 \times P_{\text{CNN}}(\text{fake})$. The ViT needed to be processed through its HuggingFace processor (applying ImageNet mean/std normalization), while the CNN received raw $[0, 255]$ pixel values as expected by its internal normalization layer. Label conventions were unified to allow both models to express $P(\text{fake})$ before averaging. This ensemble was evaluated on a 600-image balanced GAN-face dataset (300 real, 300 fake), achieving 74.00% accuracy with noticeably fewer false positives (26 FP) than either individual model.

Meta-Learner Ensemble (ViT-V1 + ViT-V2 + CNN): The second, more advanced ensemble introduced, ViT-V2, is a continual-learning variant of the ViT trained on 50,000 balanced images (CIFAKE + Manjil Karki deepfake faces) for 4 epochs with early stopping at $\text{lr}=2\text{e}-5$. A logistic regression meta-learner (“Boss”) was trained on the $P(\text{fake})$ scores from all three models (ViT-V1, ViT-V2, CNN) using a 67-image manually labeled calibration set. The meta-learner learned trust weights for each model (CNN: 0.0245, V2: 0.0392, V1: -0.0072), confirming that the CNN contributed the most reliable signal. A Random Forest analysis confirmed this, assigning feature importances of 0.46 (CNN), 0.27 (V1), and 0.26 (V2). This meta-learner ensemble was then validated on a 2,000-image blind dataset (1,000 real, 1,000 fake from the Manjil Karki set), achieving 92.25% accuracy with strong precision/recall balance (Real: 0.98 precision / 0.86 recall; Fake: 0.88 precision / 0.98 recall). An attempt to fine-tune ViT-V2 on CIPLAB and DALL-E Recognition images to improve Midjourney detection resulted in catastrophic forgetting and was not incorporated into the final ensemble.

4.7 Deployment

Both models were deployed as Gradio web apps on HuggingFace Spaces. V3 app: (<https://huggingface.co/spaces/sajaaghareeb/deepfake-detector-v3>): real/fake verdict, confidence scores, Grad-CAM overlay.

ViT app: (<https://huggingface.co/Vishwajitm01/face-ai-detector-91auc>). We did not host the ensemble model as it failed to generalize on images generated by software, which led to a suspicion that the model overfitted on select image types.

5 Experiments and Results

5.1 Quantitative Comparison

Model	Architecture	Training Data	AUC-ROC	Accuracy	Platform
V1	EfficientNetB4	CIFAKE only (~100k)	0.9924	96.00%	Google Colab
V2	EfficientNetB4	Multi-dataset (crashed)	N/A	N/A	Google Colab
V3 (Primary)	EfficientNetB4	5 datasets (~250K)	0.9751	92.20%	Lightning AI
V4 (Reverted)	EfficientNetB4	V3 + Midjourney fine-tune	N/A*	99.33% (val only)	Lightning AI
ViT-XAI	Vision Transformer	CIFAKE + 5k StyleGAN faces	0.91	91% real / 81% AI	Lightning AI
Ensemble	ViT-V1 + ViT-V2 + EfficientNetB4 (Meta-Learner)	Calibration (67) + Blind val (2,000)	N/A	92.25%	Lightning AI

*V4 AUC-ROC unavailable due to a metric logging bug. V1's higher numbers reflect CIFAKE overfitting, not true generalization. V3's lower numbers represent meaningful improvement on harder and more diverse examples. No model is perfect. V3, ViT, and the ensemble all have documented limitations.

5.2 V3 Classification Report (CIFAKE Test Set)

Class	Precision	Recall	F1-Score	Support
Real (AUTHENTIC)	0.94	0.90	0.92	10,000
Fake (AI-GENERATED)	0.90	0.95	0.92	10,000
Accuracy			0.92	20,000
Macro Average	0.92	0.92	0.92	20,000
Weighted Average	0.92	0.92	0.92	20,000

5.3 Real-World Testing — V3 HuggingFace App

Image	True Label	V3 Prediction	Confidence	Correct?
Baby in traditional outfit	Real	Real (AUTHENTIC)	94.0%	Yes
Golden retriever with a party hat	Real	Real (AUTHENTIC)	74.0%	Yes
Dog lying on grass	Real	Real (AUTHENTIC)	60.1%	Yes
Fox in field (Midjourney AI image)	AI-Generated	Real (AUTHENTIC)	52.8%	NO — False Negative
AI-generated portrait (black top)	AI-Generated	AI-Generated (FAKE)	53.0%	Yes (borderline)
Cat in astronaut suit (AI)	AI-Generated	AI-Generated (FAKE)	60.0%	Yes

The fox image (in the supplementary file) is V3's most significant failure: it is a Midjourney-generated image classified as Real (52.8%). Since V3 was never trained on Midjourney images, it lacks learned features to identify this generator's characteristic style. This is a predictable and reproducible gap, not a random error. Real photos (like the picture of the baby in traditional

clothing in the supplementary file) scored correctly, demonstrating V3's strength with authentic images. However, the model's Midjourney blind spot is a real limitation that must be addressed in future versions.

5.4 Real-World Testing — ViT-XAI Model

Image	True Label	ViT Prediction	Confidence	Correct?
Man on sofa (Midjourney)	AI-Generated	AI-Generated	63.84%	Yes
Woman in car (Midjourney)	AI-Generated	AI-Generated	80.90%	Yes
AI portrait — right of pair	AI-Generated	AI-Generated	62.16%	Yes
Real portrait — left of the pair	Real	Real	86.98%	Yes
Woman with matcha (real photo)	Real	AI-Generated	63.04%	NO — False Positive
Reddit proof-of-existence (real)	Real	AI-Generated	66.89%	NO — False Positive
Hospital birthday photo (real)	Real	AI-Generated	67.76%	NO — False Positive

The ViT model correctly identified all 3 AI-generated test images (for example, the ‘woman in car’ image in the supplementary material), but it produced false positives on 3 out of 4 real photographs (like the ‘hospital birthday’ image in the supplementary material). The model consistently cited 'High-frequency artifacts' and 'GAN/Diffusion generation patterns' for these false positives, suggesting that JPEG compression, professional retouching, or social media processing is being mistaken for generation artifacts. This calibration bias is a result of the specialist fine-tuning on StyleGAN faces: the model became oversensitized to high-quality imagery patterns.

5.5 Ensemble Model Results

Simple Ensemble (ViT + CNN, 600-image eval): On a balanced 600-image GAN-face evaluation set, the simple average ensemble matched the CNN’s overall accuracy at 74.00%, but it achieved a lower false-positive count: 26 false positives versus the CNN’s 39 and the ViT’s 74. This shows the ensemble’s ability to bring down false-positive rates by leveraging the CNN’s strength, while the ViT’s overconfidence was reduced. However, the ViT’s poor performance on this dataset (48.67% accuracy) limited the ensemble’s ceiling. The ensemble F1 was 68.55% compared to CNN's 70.11%.

Meta-Learner Ensemble (2,000-image blind validation): The logistic regression meta-learner ensemble had a 92.25% accuracy on a 2,000-image blind validation set (1,000 real, 1,000 fake from the Manjil Karki deepfake dataset). Real precision 0.98, recall 0.86, F1 0.92, and fake precision 0.88, recall 0.98, F1 0.93 showed a strong balance. This represents a considerable improvement over either individual ViT model on diverse data while maintaining competitive performance with V3. The high real-image precision (0.98) addresses the ViT’s original false-positive shortcoming, while the high fake-image recall (0.98) confirms strong detection capability. The meta-learner’s learned weights validated that the CNN was the most reliable individual contributor, consistent with its strong standalone performance across evaluation datasets.

Meta-Learner Ensemble Classification Report (2,000-image Blind Validation)

Class	Precision	Recall	F1-Score	Support
Real	0.98	0.86	0.92	1,000
Fake (AI-Generated)	0.88	0.98	0.93	1,000
Accuracy			0.92	2,000
Macro Average	0.93	0.92	0.92	2,000
Weighted Average	0.93	0.92	0.92	2,000

5.6 Head-to-Head Comparison

Dimension	V3 (EfficientNetB4)	ViT-XAI	Ensemble (Meta-Learner)
Parameters	~19M trainable (Phase 1)	~86M total	~191M combined (3 models)

Training Images	~257,283 (diverse types)	~105,000 (CIFAKE + 5k faces)	Calibration: 67; Blind val: 2,000
AUC-ROC	0.9751	0.91	N/A
Overall Accuracy	92.20% (CIFAKE test set)	~86% (real-world estimate)	92.25% (blind validation)
Midjourney Detection	FAILS — not in training data	Correct (63-81% confidence)	Failed (See Deployment 4.7)
Real-photo false positives	Low (3/3 real photos correct)	High (3/4 real photos wrong)	Low (0.98 precision on real)
Explainability	Grad-CAM (spatial saliency)	Multi-layer attention (forensic)	Meta-learner scores (no spatial maps)
Best use case	General-purpose detection	StyleGAN face forensics	Balanced general-purpose detection
Biggest limitation	Midjourney blind spot	False positives on real photos	Small calibration set; inference latency

6 Limitations and Recommendations

6.1 Known Limitations

EfficientNetB4 V3:

- V3 systematically fails on Midjourney V5/V6 images, and the fox test case confirmed this. The missing dataset is the direct cause.
- Several AI-generated images received around 50-55% confidence, indicating difficulty with highly photorealistic generation styles.
- All formal metrics were computed on CIFAKE images. Performance on other generators is likely lower.
- Training was conducted on a Lightning AI session that no longer exists. All results are from the original live run.

ViT-XAI:

- 3 of 4 real photographs were incorrectly classified as AI-generated, which is unacceptable for applications where false accusations carry reputational or legal risk.
- Optimized for StyleGAN faces, so it may not generalize to diffusion-model or Midjourney outputs.
- 86M parameters create higher inference latency compared to V3 on CPU hardware.
- The forensic factors output is template-based, not dynamically computed from the actual decision pathway.

Ensemble:

- The meta-learner was trained on only 67 manually labeled images, which risks overfitting the logistic regression weights to a narrow distribution.
- The continual-learning ViT-V2 showed high training/validation accuracy (85–87%) but only 41.79% test accuracy, showing severe catastrophic forgetting.
- Running three models sequentially increases prediction time significantly (the blind validation took over 75 minutes for 2,000 images), making real-time deployment challenging without GPU acceleration or model distillation.
- The blind validation used the same Manjil Karki dataset that ViT-V2 was partially trained on, which may boost the ensemble's reported accuracy.

6.2 Recommendations and Next Steps

- Add Midjourney training data: Sources include JourneyDB on HuggingFace and Midjourney community archives.
- Expand ensemble calibration data: The current meta-learner was trained on only 67 images. Retraining on a larger and more diverse calibration set (1,000+ images spanning multiple generators) would improve the meta-learner's weight estimates and generalization.

- Deploy ensemble as unified application: Package the three-model meta-learner ensemble as a single Gradio/HuggingFace Spaces app with individual model scores and ensemble verdict displayed in a unified dashboard.
- Confidence calibration: Apply Platt scaling or temperature scaling to reduce ViT's false-positive rate without retraining.
- Continuous retraining pipeline: New generative AI tools emerge monthly, and an automated pipeline collecting labeled samples from known AI sources would keep both models current.
- Multimodal detection: Combine visual analysis with EXIF metadata, compression signatures, and creation timestamps to improve detection on compressed social media images.
- Adversarial evaluation: Regularly test both models against adversarially crafted inputs designed to avoid detection, ensuring robustness as generators evolve.

7 Conclusion

This project developed and evaluated multiple complementary approaches to AI-generated image detection. The EfficientNetB4 V3 model, trained on ~257283 diverse images, achieved a Test AUC-ROC of 0.9751 and 92.20% accuracy, which indicates strong generalization across real-world image types. However, it failed predictably on Midjourney-generated images absent from its training set. The ViT-XAI model achieved an AUC-ROC of 0.91 with forensic-quality explainability through multi-layer attention maps. Its major flaw was producing false positives on 3 of 4 real photographs tested. To address these complementary failure modes, we developed a meta-learner ensemble combining ViT-V1, ViT-V2, and EfficientNetB4, which achieved 92.25% accuracy on a 2,000-image blind validation with strong precision/recall balance (Real: 0.98/0.86; Fake: 0.88/0.98). While no single model or ensemble is production-ready without further development, the ensemble demonstrates that multi-architecture combinations can mitigate individual model weaknesses, and we have documented all limitations explicitly.

The progression from V1 (CIFAKE-only, 96%) to V3 (multi-dataset, 92.2%) shows that lower accuracy on a narrow benchmark can reflect genuine improvement when the model learns from harder and more diverse examples. The V4 catastrophic forgetting experiment demonstrated that narrow fine-tuning on a small specialist dataset can destroy broad generalization, and this is a finding corroborated by the ViT-V2 continual learning attempt that achieved 85–87% validation accuracy but only 41.79% on independent test data. Both cases serve as cautionary examples for any team considering incremental updates to deployed models. The V2 crash on Google Colab motivated the migration to Lightning AI, emphasizing the importance of platform selection for large-scale training runs.

Social and Managerial Implications: False positives, meaning incorrectly flagging real photos as AI-generated, can cause reputational damage and reduce trust in detection systems. Our ViT results demonstrated this risk, while the ensemble's 0.98 precision on real images shows that multi-model approaches can substantially mitigate it. Detection systems must be evaluated not only by aggregate accuracy but by the different error rates for all image types and demographic groups. From a managerial perspective, organizations that want to deploy these tools must establish human-in-the-loop escalation protocols for borderline cases. The dual-use nature of deepfake detection (the same knowledge that enables detection can accelerate evasion) highlights the importance of responsible disclosure, transparent benchmarking, and ongoing adversarial evaluation.

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